EILENBERG'S SWINDLE: A MODULE WHOSE SUM WITH A NONFINITELY GENERATED FREE MODULE IS FREE IS ITSELF ABSORBED BY THAT FREE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 38970-38985 (canonical tag 0GVF) and the counted proof is the proof environment at source lines 38987-38993, which immediately follows it and is its own proof.

Editorial note. The counted statement contains three of the author's special environments: the slogan recording the name of the result, the history recording an anecdote with a citation, and the reference recording the source of the statement. The retained preamble defines all three as comment-type, so their bodies are discarded when the document is typeset and the two citations they carry are never evaluated; this is why the card needs no bibliography, and the environments are kept verbatim inside the counted span.

Editorial note. The proof is a single displayed chain of isomorphisms. It is deliberately short but complete: the chain itself is the swindle and no step is left to the reader, so the card is not an instance of an omitted argument.

Editorial note. The counted span contains no cross reference, so there is no navigation stub; the preamble span runs to line 90, the current end of the retained preamble, which is exactly what makes the three environments comments.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key and by this note.

Allowed standard prerequisites (editorial).

- A nonfinitely generated free module is isomorphic to a countably infinite direct sum of itself, so its direct summands can be shuffled without changing the isomorphism type. used at proof lines 38988-38992. Produces the middle steps of the displayed chain of isomorphisms, which is the whole argument.
- If a free module is isomorphic to the direct sum of two modules, then summing with one of them again yields the same free module. used at proof lines 38988-38992. Justifies the last step of the chain and hence the statement.

Lemma 0.2 (Eilenberg's lemma)**.** *If $P \oplus Q \cong F$ with F a nonfinitely generated free module, then $P \oplus F \cong F$.*

Proof.

$$F \cong F \oplus F \oplus \ldots \cong P \oplus Q \oplus P \oplus Q \oplus \ldots \cong P \oplus F \oplus F \oplus \ldots \cong P \oplus F$$

□

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